

The Missing Band-Indicator Wavelets Are Frames

A candidate full solution to Problem 2.7 of arXiv:1905.08230

Autonomous proof candidate for human review

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Status. Candidate full affirmative solution, likely valid. For every $1/8 < b \leq 1/6$, the function whose Fourier transform is $\mathbf{1}_{(-1,-b) \cup (b,1)}$ is a dyadic frame wavelet. The proof gives the explicit non-optimal frame bounds $1/54$ and 6 under the Fourier convention below.

1 The question and answer

Bownik and Rzeszotnik ask the following question in Problem 2.7 of [1]. For $0 < b < 1$, define $\psi_b \in L^2(\mathbb{R})$ by

$$\widehat{\psi}_b = \mathbf{1}_{E_b}, \quad E_b = (-1, -b) \cup (b, 1).$$

For which $1/8 < b \leq 1/6$ is ψ_b a frame wavelet? The neighboring parameter ranges were determined by Bownik and Weber [2], who also expressed heuristic confidence that the missing interval should be positive.

in [24]. Despite partial progress in a subsequent work of Christensen et al. [25], either formulation of this problem remains open. It is worth adding that an analogue of Problem 2.6 for Gabor Bessel sequences has been proven in [24, Theorem 3.1].

2.7. A simple question that nobody has bothered to answer. The last problem illustrates the difficulty of determining whether a function is a frame wavelet or not. The following problem was proposed by Weber and the first author [21].

Problem 2.7. For $0 < b < 1$ define $\psi_b \in L^2(\mathbb{R})$ by $\widehat{\psi}_b = \mathbf{1}_{(-1,-b) \cup (b,1)}$. For what values of $1/8 < b \leq 1/6$, is ψ_b a frame wavelet?

The above range of parameter b seems to be the hardest in determining a frame wavelet property of ψ_b . Outside of this range, the following table lists properties of ψ_b which were shown in [21].

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Figure 1: Problem 2.7 in the source survey, PDF page 7.

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \psi_{j,k}(x) = 2^{j/2} \psi(2^j x - k).$$

Theorem 1. For every $1/8 < b \leq 1/6$ and every $f \in L^2(\mathbb{R})$,

$$\frac{1}{54} \|f\|_2^2 \leq \sum_{j,k \in \mathbb{Z}} |\langle f, (\psi_b)_{j,k} \rangle|^2 \leq 6 \|f\|_2^2. \quad (1)$$

Consequently every parameter in the interval asked about in Problem 2.7 gives a frame wavelet.

2 Exact periodization and the alias graph

Put $F = \widehat{f}$. At a fixed scale $P = 2^j$, Parseval's identity for Fourier series on $[0, P)$ gives

$$\sum_{k \in \mathbb{Z}} |\langle f, (\psi_b)_{j,k} \rangle|^2 = \int_0^P \left| \sum_{n \in \mathbb{Z}} F(x + nP) \mathbf{1}_{PE_b}(x + nP) \right|^2 dx. \quad (2)$$

Since $PE_b \subset (-P, P)$, only $n = 0, -1$ can occur. Hence the right-hand side is exactly

$$\int_0^{bP} |F(x - P)|^2 dx + \int_{bP}^{(1-b)P} |F(x) + F(x - P)|^2 dx + \int_{(1-b)P}^P |F(x)|^2 dx. \quad (3)$$

This formula has a useful graph interpretation. Take two signed copies of the positive half-line,

$$X = \{+, -\} \times (0, \infty), \quad d\mu(\sigma, t) = dt,$$

and define

$$g(+, t) = F(t), \quad g(-, t) = -F(-t).$$

For every dyadic power P , join $(+, t)$ to $(-, P - t)$ whenever

$$bP < t < (1 - b)P. \quad (4)$$

At either signed vertex of magnitude t , put one *pin* whenever

$$(1 - b)P < t < P. \quad (5)$$

Edges carry Lebesgue measure in t ; reflection $t \mapsto P - t$ preserves that measure. Summing (3) over j yields

$$\sum_{j,k \in \mathbb{Z}} |\langle f, (\psi_b)_{j,k} \rangle|^2 = \mathcal{E}(g) := \int_{v \sim w} |g(v) - g(w)|^2 d\nu + \int_{v \text{ pinned}} |g(v)|^2 d\mu. \quad (6)$$

The sign change on the negative copy is the reason that the middle term in (3) becomes a difference across an edge.

3 The graph has uniform measured expansion

All statements below ignore countably many dyadic boundary points, which form a μ -null set. Write a magnitude uniquely as

$$t = 2^m r, \quad m \in \mathbb{Z}, \quad 1 < r < 2, \quad (7)$$

and set

$$c = 8b \in (1, 4/3], \quad a = 2(1 - b) \in [5/3, 7/4).$$

Lemma 1. *The total incidence of a vertex (graph edges plus pins) is as follows:*

<i>mantissa</i>	<i>graph degree</i>	<i>number of pins</i>
$1 < r < c$	2	0
$c < r < a$	3	0
$a < r < 2$	2	1.

Moreover, no two vertices in the first row are adjacent.

Proof. For a vertex of magnitude $2^m r$, the only possible active powers are $2^{m+1}, 2^{m+2}, 2^{m+3}$. Indeed, the first two are always active, the third is active exactly when $r > 8b = c$, and all later powers are inactive because $16b > 2 > r$. The 2^{m+1} incidence is an edge for $r < a$ and a pin for $r > a$; all other active incidences are edges. This proves the table.

If $1 < r < c$, its neighbors along the first two active powers have dyadic mantissas

$$4 - 2r \quad \text{and} \quad 2 - r/2. \quad (8)$$

Since $c \leq 4/3$, both quantities are strictly larger than c when $r < c$. Thus the two-edge, unpinned class is independent. $\square$

Orient each graph edge from the endpoint of larger magnitude toward the endpoint of smaller magnitude. A vertex of magnitude t has at most one outgoing edge. In fact, if $P = t + s$ and $s < t$, then the overlap condition at t implies

$$\frac{t}{1-b} < P < 2t. \quad (9)$$

The ratio of the endpoints in this interval is $2(1-b) < 2$, so it contains at most one power of two.

Let $A \subset X$ be measurable with finite measure. Write

$$\ell = \mu(A \cap L), \quad h = \mu(A \cap H),$$

where L is the first row of Lemma 1 and $H = X \setminus L$. Let $B(A)$ be the measure of graph edges crossing from A to $X \setminus A$, plus the measure of pins based in A . Finally, let $e(A)$ denote the measure of graph edges having both endpoints in A .

Because the edge orientation has out-degree at most one and each reflection preserves measure,

$$e(A) \leq \mu(A) = \ell + h. \quad (10)$$

The incidence table and the handshake identity now give

$$B(A) = 2\ell + 3h - 2e(A) \geq h. \quad (11)$$

Every L -vertex has two graph neighbors, both in H . Counting these incidences inside A , and charging the rest to crossing edges, gives

$$2\ell \leq 3h + B(A). \quad (12)$$

If $\ell \leq 2h$, then (11) implies $B(A) \geq (\ell + h)/3$. If $\ell > 2h$, then (12) implies the same inequality. We have proved the measured isoperimetric estimate

$$B(A) \geq \frac{1}{3}\mu(A). \quad (13)$$

4 Cheeger inequality and proof of the theorem

First take g bounded with support of finite measure and put $A_s = \{v : |g(v)|^2 > s\}$. Integrating (13) over $s > 0$ and using the layer-cake formula gives

$$\frac{1}{3}\|g\|_2^2 \leq \int_{v \sim w} ||g(v)|^2 - |g(w)|^2|, d\nu + \int_{v \text{ pinned}} |g(v)|^2, d\mu. \quad (14)$$

By Cauchy–Schwarz and the elementary inequality $||z| - |w|| \leq |z - w|$, the right-hand side is at most

$$\mathcal{E}(g)^{1/2} \left(\int_{v \sim w} (|g(v)| + |g(w)|)^2 d\nu + \int_{v \text{ pinned}} |g(v)|^2 d\mu \right)^{1/2}. \quad (15)$$

The second factor squared is bounded by $6\|g\|_2^2$: each graph edge is charged twice at each endpoint after using $(u + v)^2 \leq 2(u^2 + v^2)$, and Lemma 1 gives

$$2 \deg_{\text{graph}}(v) + \deg_{\text{pin}}(v) \leq 6.$$

Therefore

$$\mathcal{E}(g) \geq \frac{1}{54}\|g\|_2^2.$$

Truncation and density extend this estimate to every $g \in L^2(X)$.

Conversely, $|z - w|^2 \leq 2(|z|^2 + |w|^2)$ and the same incidence table give

$$\mathcal{E}(g) \leq 6\|g\|_2^2.$$

Since (6) is exact and Plancherel gives $\|g\|_{L^2(X)} = \|F\|_2 = \|f\|_2$, these are precisely the two inequalities in (1). This proves Theorem 1.

5 Endpoint and sharp-threshold checks

At $b = 1/6$, the exceptional mantissa $r = 4/3$ supports a degree-two cancellation chain. It is a countable null fiber, so it does not affect the L^2 estimate. On either side, Lemma 1 forces the extra branching used by (13).

The mechanism breaks at exactly the known negative threshold. If $b > 1/6$, then $c = 8b > 4/3$, and the two quantities in (8) can again lie below c . Adjacent unpinned degree-two vertices can then form arbitrarily long cancellation corridors, in agreement with the construction in [2]. Thus the graph proof also explains why $1/6$ is the endpoint.

6 Novelty status

A bounded search of the run indexes, exact phrases from Problem 2.7, the source survey, the original Bownik–Weber paper, and later web/arXiv records located no published solution of the missing interval. Priority is not asserted. The result should be checked by a wavelet/frame specialist, especially the measured edge-counting formulation, before dissemination.

References

- [1] M. Bownik and Z. Rzeszotnik, *Open problems in wavelet theory*, in *Operator Theory, Operator Algebras and Their Interactions with Geometry and Topology*, Oper. Theory Adv. Appl. 278 (2020), 77–100; arXiv:1905.08230.
- [2] M. Bownik and E. Weber, *Affine frames, GMRA's, and the canonical dual*, Studia Math. 159 (2003), 453–479.